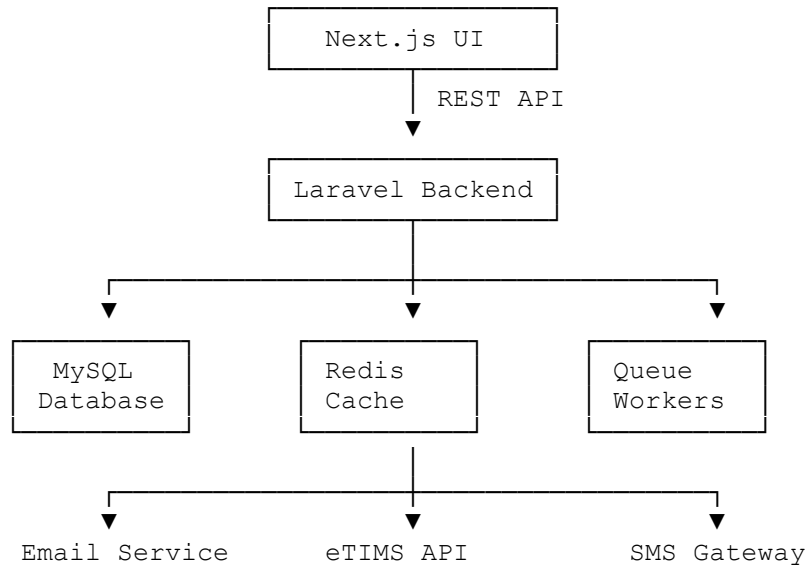


YADA CRM SYSTEM DESIGN

High-Level System Design



Core Modules

1. Authentication & RBAC

Tables

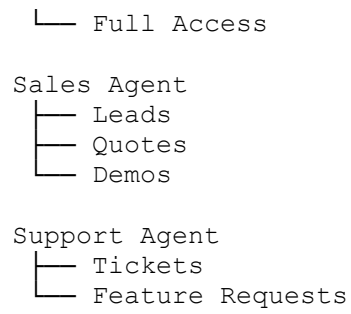
```
users
roles
permissions
role_user
permission_role
```

Flow

```
User Login
↓
JWT/Sanctum Token
↓
Check Role
↓
Check Permission
↓
Grant Access
```

Example:

```
Admin
├── Manage Users
└── Manage Roles
```



2. Lead & Sales Pipeline

Tables

leads
quotes
demos
technical_reviews
deals
tasks

Pipeline

Lead
↓
Quote
↓
Demo
↓
Technical Review
↓
Closed Won

Example

ABC Bank
↓
Lead
↓
Quoted KES 500,000
↓
Demo Completed
↓
Technical Review Passed
↓
Closed

3. Subscription Management

Tables

clients

subscription_plans
subscriptions
licenses
renewals

Flow

Client
↓
Choose Plan
↓
Subscription Created
↓
Track Usage
↓
Renew

Example:

Client: XYZ Ltd

Plan:
Enterprise

Seats:
50

Used:
32

4. Communication Hub

Tables

emails
sms_logs
meeting_notes
activities
templates

Unified Timeline

Client Record

10:00 Email Sent
11:00 SMS Sent
12:00 Demo Conducted
14:00 Meeting Notes Added

Everything appears in one timeline.

5. Support Module

Tables

tickets
ticket_comments
feature_requests
attachments

Workflow

Open
↓
Assigned
↓
In Progress
↓
Resolved
↓
Closed

6. Analytics Dashboard

Metrics

Revenue

MRR
ARR
Total Revenue

Sales

Lead Conversion %
Quote Conversion %
Demo Success %

Customer Success

Renewal Rate
Churn Rate
Active Users

7. Invoice & eTIMS Integration

Tables

invoices
invoice_items
payments
etims_logs

Flow

Invoice Generated
↓
Send to eTIMS
↓
Receive eTIMS Code
↓
Store Response

Database Design

users
roles
permissions

clients

leads
quotes
demos
deals

subscriptions
plans
licenses

tickets
feature_requests

emails
sms_logs
activities

invoices
payments

audit_logs

All tables should use:

UUID Primary Key

Example:

```
id CHAR(36) PRIMARY KEY
```

Laravel Queues

Use Laravel Queues for:

Renewal Reminders

Daily Cron
↓
Check Expiring Subscriptions

↓
30 Days Remaining?
↓
Send Email

Other Jobs

Email Sending
SMS Sending
Invoice Generation
eTIMS Sync
Analytics Calculation

API Design

/api/auth

/api/users
/api/roles

/api/leads
/api/quotes
/api/deals

/api/clients

/api/subscriptions
/api/plans

/api/tickets

/api/invoices

/api/dashboard

Deployment Architecture

Internet

↓
Nginx

↓
Laravel API

↓
MySQL

Next.js Frontend
↓
Vercel

Redis
↓
Caching + Queues

